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## **Securing Wireless Vehicle Communications**

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## **Abstract**

Vehicular Ad-Hoc Networks (VANETs) are critical for modern intelligent transportation systems, enabling real-time communication between vehicles and infrastructure. However, securing vehicular communications against threats such as Sybil attacks, replay attacks, and trust score manipulation remains a significant challenge. This dissertation proposes the Hybrid Reputation-Based Blockchain & Proxy Vehicle Authentication Model (HB-RPVAM), integrating reputation-based trust mechanisms, proxy vehicle batch authentication (ID-MAP), and blockchain-based smart contract storage. To enhance security and efficiency, the model introduces multi-signature verification for proxy authentication, Layer 2 blockchain scaling, AI-based proxy selection, and behaviour-based trust scoring. The proposed framework significantly reduces computational complexity and communication overhead and complexity while ensuring low-latency authentication and secure trust score updates. The system is validated using LabVIEW simulations, demonstrating its scalability and resilience against malicious attacks.

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# 1. Introduction

Vehicular Ad Hoc Networks (VANETs) are a cornerstone of Intelligent Transportation Systems (ITS), enabling vehicles to communicate with each other (Vehicle-to-Vehicle, V2V) and with infrastructure (Vehicle-to-Infrastructure, V2I) [1]. These communications facilitate various critical applications such as collision avoidance, traffic congestion management, automated driving systems, emergency warning systems, and route optimization [2]. The integrity, reliability, and security of these communications directly impact passenger safety, vehicular efficiency, and overall road safety. Thus, ensuring secure, trustworthy, and efficient authentication mechanisms in vehicular communications is paramount.

Despite their importance, secure vehicular communication systems face significant threats. Attackers can intercept, manipulate, or fabricate communication messages, leading to catastrophic outcomes such as accidents, traffic disruptions, or severe breaches of privacy [2], [3]. Thus, establishing a robust and secure authentication mechanism is vital to guarantee that only genuine and trustworthy vehicles can exchange valid information.

As connected and autonomous vehicles (CAVs) become increasingly prevalent, the volume of authentication requests dramatically escalates [4]. Conventional authentication methods must adapt to handle not only security requirements but also the massive scale of connected devices, real-time data exchange constraints, and evolving cyber threats. Therefore, there's a pressing need for advanced authentication frameworks that balance strong security measures with computational efficiency and minimal latency.

## 1.1 Importance of Secure Vehicular Authentication

Secure vehicular authentication is critical because it addresses three main challenges [5]:

1. **Security and Privacy:** Ensuring message integrity, authenticity, and non-repudiation are paramount to avoid malicious attacks and preserve user privacy. Unsecured vehicular communications could expose vehicles to attacks such as impersonation, replay, Sybil, and Man-in-the-Middle (MITM) attacks [6], [7].
2. **Real-Time Performance:** Vehicle networks require rapid authentication processes. Delays of even milliseconds in verifying a vehicle's authenticity could result in fatal incidents, especially at high speeds [2], [8]. Thus, the authentication system must be designed for minimal latency and high throughput.



3. **Scalability and Efficiency:** Authentication schemes must handle potentially thousands of simultaneous authentication requests without compromising speed or security [9], [10]. Efficient computational algorithms and decentralized mechanisms are necessary to prevent bottlenecks and delays inherent in centralized authentication systems.

Given these stringent requirements, traditional authentication methods struggle significantly in meeting the demands of modern vehicular networks. Therefore, it is essential to explore innovative methods to achieve secure, efficient, and scalable vehicular authentication.

## 1.2 Research Problem Statement

Despite numerous existing authentication frameworks, they often have intrinsic shortcomings that limit their effectiveness in real-world scenarios. The primary existing authentication methodologies include:

- **Public Key Infrastructure (PKI)-Based Authentication:** PKI schemes have historically been widely used to secure communications in vehicular networks. PKI relies heavily on digital certificates and a centralized Certificate Authority (CA) to validate identities. However, certificate validation and revocation list management introduce significant computational overhead and latency [2], [3], [10], making PKI highly resource-intensive, particularly in large-scale, highly dynamic networks. Additionally, PKI schemes experience scalability challenges as the number of vehicles grows exponentially.
- **Blockchain-Based Authentication:** Blockchain technology offers decentralized, immutable trust management solutions. However, current blockchain-based authentication mechanisms face significant scalability challenges due to high transaction latency and expensive gas costs associated with blockchain operations [4], [11]. Storing every authentication event on-chain is impractical for large-scale, dynamic vehicular networks where thousands of vehicles simultaneously authenticate.
- **ID-Based Proxy Authentication (ID-MAP):** ID-MAP proxy authentication significantly reduces the computational overhead by offloading verification tasks from RSUs to selected proxy vehicles, performing batch authentication. However, ID-MAP lacks a comprehensive trust mechanism to differentiate between trustworthy and malicious proxy vehicles [10], [12], making it susceptible to attacks such as proxy compromise and collusion.

Given these inherent limitations, the research community urgently requires a novel authentication model that effectively mitigates these issues, providing security, scalability, and computational efficiency simultaneously.

### 1.3 Proposed Research Contributions

This dissertation proposes a novel Hybrid Reputation-Based Blockchain & Proxy Vehicle Authentication Model (HB-RPVAM) designed explicitly to address the shortcomings identified in existing schemes. HB-RPVAM achieves a holistic improvement by combining state-of-the-art methods from three distinct authentication domains:

- **Reputation-Based Authentication:** This mechanism dynamically evaluates and updates the trustworthiness of vehicles based on their historical interactions and current behaviors [13], [14], [15]. Reputation scores help to identify malicious or compromised vehicles swiftly, enabling proactive security measures.
- **Proxy Vehicle-Based Batch Authentication (ID-MAP Integration):** By employing proxy vehicles for batch verification, the proposed model significantly reduces computational overhead on Roadside Units (RSUs). The batch verification mechanism aggregates multiple authentication requests into a single computational step [16], lowering complexity from linear ( $O(n)$ ) to logarithmic ( $O(\log n)$ ) [10], [17], [18].
- **Blockchain-Based Smart Contract Storage:** Blockchain technology is integrated to securely manage and store reputation scores and critical authentication records. This decentralized approach ensures immutability, transparency, and resistance to data tampering and single-point failures [4], [11].
- **AI-Based Trust Scoring & Multi-Signature Proxy Verification:** To counteract vulnerabilities such as compromised proxies and collusion attacks, HB-RPVAM introduces multi-signature verification, where multiple proxies or RSUs must independently verify and confirm authentication requests. Additionally, an AI-based trust scoring system is implemented to monitor real-time vehicle behaviors, dynamically updating trust levels based on detected anomalies in driving patterns [13], [19].

In summary, the novel contributions of this dissertation are clearly described as:

- Development of an integrated hybrid authentication framework incorporating reputation-based trust management with blockchain decentralization and ID-MAP proxy authentication mechanisms.
- Introduction of AI-driven adaptive trust scoring for dynamic reputation updates and enhanced security.
- Implementation of multi-signature verification to prevent single-point failures and proxy vehicle compromise.
- Optimization of blockchain interactions using Layer 2 scaling techniques to dramatically improve scalability and reduce latency.

The HB-RPVAM framework not only addresses the limitations identified in existing authentication methods but also sets a solid foundation for future research and real-world applications, ultimately providing secure, efficient, and robust authentication tailored specifically for vehicular communication environments.

## 2. State of the Art

Modern vehicular ad-hoc networks (VANETs) typically rely on three major classes of authentication methods: Public Key Infrastructure (PKI), blockchain-based authentication, and identity-based proxy approaches (e.g., ID-MAP). Each has distinct mechanisms, computational costs, and vulnerabilities. This section presents how these methods work, highlighting their core strengths as well as key limitations.

### 2.1 PKI-Based Authentication

A Public Key Infrastructure (PKI) establishes trust through a Certificate Authority (CA) that issues digital certificates [2], [20]. Each vehicle  $V_i$  stores a private key  $sk_i$  and a certified public key  $pk_i$ . The CA's signature on  $pk_i$  attests to  $V_i$ 's legitimacy [3]. When  $V_i$  sends a message  $m_i$ , it generates a digital signature

$$\sigma_i = \text{Sign}_{sk_i}(m_i) \quad (2.1)$$

and includes  $\sigma_i$  along with its certificate  $Cert_i$  in the broadcast. Receivers verify:

- The CA's signature on  $Cert_i$  (proving that  $pk_i$  is legitimate).
- The sender's signature  $\sigma_i$  on  $m_i$  using  $pk_i$ .

$$Verify_{pk_i}(m_i, \sigma_i) \rightarrow \{\text{True}, \text{False}\} \quad (2.2)$$

Here,  $pk_i$  is retrieved from the certificate, which itself must be up to date (i.e., not revoked).

### Limitations.

- **High Overhead for Certificate Management:** With potentially thousands of vehicles in range, each incoming message has an accompanying certificate to be checked [21]. This inflates both communication and computation.
- **Revocation Complexity:** A compromised vehicle's certificate must appear on a Certificate Revocation List (CRL), and every receiver must consult large CRLs in real time [2], [3]. The verification time can grow with the number of revoked certificates, limiting scalability.
- **Latency Concerns:** For dense traffic, an RSU verifying many certificates can face significant delays [21].

Formally, if  $n$  vehicles broadcast messages nearly simultaneously, each verification includes checking certificate validity  $T_{cert}(n)$  and checking signature correctness  $T_{sig}(n)$ . Hence, the combined time roughly scales as:

$$T_{PKI} = n(T_{cert} + T_{sig}) \quad (2.3)$$

making PKI quickly unwieldy in high-density VANET scenarios.

## 2.2 Blockchain-Based Authentication

Blockchain-based authentication replaces the centralized CA with a decentralized ledger of authenticity [2], [4], [11], where each vehicle or RSU submits relevant identity or trust updates on-chain. A blockchain network maintains an immutable log of events (transactions) that help nodes verify that a particular vehicle is legitimate or that its key material has not been revoked.

- **Identity/Key Registration** - Vehicles register their pseudonyms  $PID_i$  or public keys on-chain [1], often via transactions:

$$Tx_i = H(PID_i \parallel m_i \parallel t_i) \quad (2.4)$$

where  $H(\cdot)$  is a cryptographic hash function, and  $m_i$  might be any metadata or certificate info.

- **Verification** - Receivers consult the blockchain state to confirm that the sender's identity is valid (not revoked, not malicious) [4]. Because the blockchain is shared, many checks (e.g., revocation) do not rely on a single CA.

### Limitations.

- **Transaction Latency & Gas Costs:** Each update or revocation typically requires an on-chain transaction (e.g., on Ethereum) [2]. This becomes too costly and slow when  $n$  vehicles frequently update their states.
- **Throughput Constraints:** Public blockchains often support limited transactions per second, creating network bottlenecks in high-mobility VANETs [11].
- **Synchronization Delays:** Confirming a transaction can take seconds or even minutes, whereas vehicle safety messages need to be processed within milliseconds [4].

Even if using faster private or consortium blockchains, large-scale VANET with thousands of vehicles still faces overhead from repeated on-chain writes. If each authentication requires:

$$GasCost(Tx_i) = GasBase + GasData(m_i) \quad (2.5)$$

then for  $n$  vehicles, total cost grows linearly:  $n \times GasCost(Tx_i)$ . This is untenable for real-time safety-critical tasks, making naive on-chain solutions impractical without significant off-chain or Layer-2 optimizations.

### 2.3 ID-MAP (Identity-Based Proxy Authentication)

Unlike PKI or blockchain-centric approaches, Identity-Based Message Authentication with Proxy Vehicles (ID-MAP) delegates most verification tasks to designated proxy vehicles [10]. Instead of having RSUs individually authenticate every signature, ID-MAP selects a proxy that batch-verifies multiple vehicles' signatures at once, drastically reducing the RSU's workload.

A trusted authority issues identity-based keys  $(sk_i, pk_i)$  to each vehicle  $V_i$ . These can be derived from a master secret [10], ensuring that vehicles can sign messages without needing separate certificates.

- **Message Signing** - If  $V_i$  wants to broadcast data  $m_i$  at time  $t_i$ , it computes a signature (e.g., via elliptic curve ElGamal or similar):

$$\sigma_i = \text{Sign}_{\text{sk}_i}(m_i, t_i) \quad (2.6)$$

- **Proxy Vehicle Batch Verification** - In ID-MAP, a *proxy vehicle*  $P_v$  collects messages  $\{m_1, \dots, m_d\}$  from  $d$  different senders. It aggregates their signatures into a single check:

$$\sum_{i=1}^d \sigma_i \leftrightarrow \sum_{i=1}^d (pk_i \text{ or other sign parameters}). \quad (2.7)$$

If the aggregated equality holds,  $P_v$  concludes all  $d$  messages are valid. Then  $P_v$  forwards a single “batched authentication result” to the RSU [17], [18].

### Limitations.

- **Malicious Proxy Risk:** If the chosen proxy  $P_v$  is compromised, it can falsely approve invalid signatures [12] or fail to detect manipulated messages. Without a robust reputation or trust check, a single bad proxy can undermine the scheme for an entire batch.
- **No Inherent Trust Differentiation:** Classical ID-MAP does not incorporate dynamic reputation or distributed consensus to confirm the proxy’s honesty [8], [10]. If  $P_v$  is randomly picked, there is a non-negligible probability of selecting a malicious node as the proxy, especially in crowded networks.

Formally, the batch verification equation for  $d$  messages might look like:

$$\left( \sum_{i=1}^d s_i \right) P = \sum_{i=1}^d (h(m_i)Q_i) + \dots \quad (2.8)$$

where  $s_i$  are each vehicle’s partial signatures,  $Q_i$  are public keys or ephemeral parameters, and  $P$  is a generator in the elliptic curve group. A proxy can claim correctness even if part of the batch is invalid thus if the proxy is malicious and passes the entire batch, other vehicles or the RSU might accept them all erroneously.

### **3. Proposed HB-RPVAM Model**

#### **3.1 System Modelling**

The network comprises the following entities;

##### ***3.1.1 Trusted Authority (TA)***

The Trusted Authority (TA) is a central entity initially responsible for vehicle registration, initialization of cryptographic parameters, and assignment of initial reputation scores [2], [6], [10]. In HB-RPVAM, the TA issues unique identity credentials and cryptographic keys to vehicles upon their entry into the network [6], [21]. It also manages initial security credentials and reputation score initialization, ensuring that each vehicle enters the network with a fair and secure baseline score. After this initial setup, the TA's role decreases significantly, transitioning to decentralized management via blockchain and proxy vehicles [4]. Thus, while initially centralized, the TA does not represent an ongoing single point of failure, as decentralized authentication is subsequently employed.

##### ***3.1.2 Proxy Vehicles (PVs)***

Proxy Vehicles (PVs) are dynamically selected vehicles tasked with verifying authentication requests on behalf of Roadside Units (RSUs). These vehicles significantly reduce computational overhead using batch signature verification [18]. In HB-RPVAM, proxy vehicles aggregate multiple authentication requests and perform batch verification, drastically improving computational efficiency. PVs use Elliptic Curve Digital Signature Algorithm (ECDSA) for cryptographic verification of messages [22], [23]. Critically, HB-RPVAM employs an AI-based trust-scoring mechanism to ensure only high-reputation vehicles become proxies. Additionally, multiple independent proxy vehicles must perform multi-signature verification, significantly mitigating risks of compromise or collusion [12].

##### ***3.1.3 Roadside Units (RSUs)***

Roadside Units (RSUs) are fixed infrastructural components that facilitate vehicular communication. They serve as a critical point in authenticating proxy vehicle batch verification results before recording reputation updates onto the blockchain [9]. RSUs ensure correctness and integrity of the proxy batch verification process by verifying multi-signatures provided by PVs [18]. They also manage secure communication with vehicles, update trust scores, and ensure that authentication results are accurately reflected on the blockchain [2]. While proxy

vehicles perform initial batch verification [16], RSUs provide the critical second layer of verification required by HB-RPVAM's multi-layered authentication approach.

#### ***3.1.4 Onboard Units (OBUs)***

Each vehicle is equipped with an Onboard Unit (OBU), a specialized computational device responsible for generating cryptographic keys, signatures, and authentication requests [3], [6]. OBUs generate Elliptic Curve Cryptographic (ECC) based key pairs, perform digital signatures (ECDSA), store their own private keys securely, and maintain dynamic reputation scores [22], [24]. They initiate authentication requests, engage in batch verification with proxy vehicles, and communicate securely with RSUs and blockchain smart contracts. OBUs play a central role in generating trust information, cryptographic credentials, and behavioural data utilized by HB-RPVAM's adaptive authentication mechanisms.

#### ***3.1.5 Smart Contract-Based Blockchain***

A smart contract-based blockchain serves as the decentralized backbone for secure and immutable reputation score storage and updates [11]. The blockchain smart contract maintains secure records of each vehicle's dynamically updated reputation score and authentication results, ensuring transparency, immutability, and decentralized verification [2], [4]. HB-RPVAM leverages blockchain technology primarily for storing reputation updates securely, significantly reducing the frequency and size of on-chain transactions through Layer 2 scaling solutions. Only reputation updates and critical authentication events are stored permanently on the blockchain, reducing gas costs, transaction latency, and enhancing scalability.

#### ***3.1.6 Layer 2 Blockchain Solutions***

To address the blockchain's inherent latency and high transaction costs, HB-RPVAM integrates Layer 2 scaling solutions, such as Rollups or sidechains [2]. Layer 2 solutions facilitate processing most authentication activities off-chain, significantly decreasing the blockchain load. Authentication requests validated by proxy vehicles are processed off-chain, while periodic reputation updates are batched and stored securely on-chain via smart contracts. This dramatically improves authentication response times, allowing vehicles to authenticate swiftly without sacrificing blockchain's security benefits.



### 3.1.7 AI-Based Proxy Selection and Anomaly Detection

HB-RPVAM employs artificial intelligence (AI) to dynamically select proxy vehicles based on trust scores and real-time driving behaviors [25]. An AI-based trust scoring system analyses historical and real-time behavioural data, such as braking frequency, speed consistency, acceleration patterns, route adherence, and other driving metrics [14], [15]. Vehicles exhibiting consistent behaviour receive higher reputation scores, qualifying them to become proxy vehicles. Additionally, anomaly detection algorithms identify abnormal or suspicious behaviour [26], quickly downgrading reputation scores, and ensuring the integrity of the authentication process. AI integration prevents malicious or compromised vehicles from infiltrating proxy selection [13], [19], significantly enhancing overall security.

## 3.2 Scheme Modelling

The proposed Hybrid Reputation-Based Blockchain & Proxy Vehicle Authentication Model (HB-RPVAM) unfolds in four primary phases (see Figure 1): Vehicle Registration, Proxy Vehicle-Based Batch Authentication, Blockchain-Based Trust Management, and Adaptive Authentication With AI-Based Security.

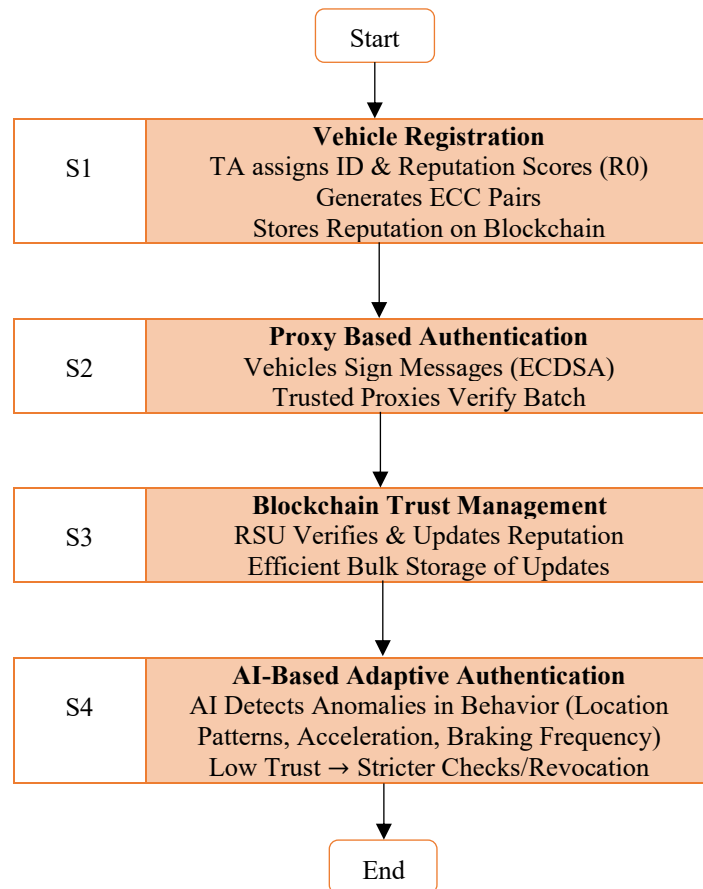


Figure 1: Flowchart of the proposed HB-RPVAM scheme

## Phase 1: Vehicle Registration

### *Step 1 - Initial Setup by the Trusted Authority (TA)*

- The TA generates global system parameters and a master secret key.
- Each vehicle  $V_i$  registers with the TA and obtains a unique identity  $ID_i$  (or pseudonym) and an initial reputation score  $R_i^0$ , reflecting how trustworthy  $V_i$  is at the outset.

### *Step 2 - ECC-Based Key Generation*

- Each vehicle's onboard unit (OBU) runs an elliptic curve cryptography (ECC) process to produce its private and public keys:

$$x_i \in \mathbb{Z}_q, \quad P_i = x_i \cdot G \quad (3.1)$$

where  $G$  is the chosen base point (generator) on the elliptic curve and  $q$  is the curve's order. The private key  $x_i$  remains securely stored in the OBU, whereas  $P_i$  can be shared or used for verification.

### *Step 3 - Blockchain/Contract Initialization*

- The TA (or an RSU operating under TA's authority) record's each vehicle's initial reputation  $R_i^0$  in a blockchain-based smart contract, ensuring an immutable reference for future updates.

$$R_i^0 = \alpha \times \frac{1}{n} \sum_{j=1}^n f_j \quad (3.2)$$

Initial reputation is calculated by taking an average of collected feedback values  $\{f_j\}$ , then scaling the average by a normalization factor  $\alpha$  where  $n$  is the number of feedback sources or separate data points considered. At the end of this phase, every vehicle is recognized by the TA, has an initial reputation score stored on-chain, and possesses ECC key pairs for signing messages.

## Phase 2: Proxy Vehicle-Based Batch Authentication

In conventional ID-MAP, a single proxy verifies multiple messages in batch. Here, we enhance the process by selecting reputable vehicles as proxies and, if needed, requiring more than one proxy to mitigate insider threats.

### *Step 1 - Signing a Message*

- Whenever  $V_i$  needs to broadcast a message  $m_i$  at time  $t_i$ , it generates a digital signature via ECDSA:

$$r_i = (k_i \times G)_x \mod q, \quad s_i = k_i^{-1}(H(m_i) + x_i r_i) \mod q \quad (3.3)$$

where  $k_i \in \mathbb{Z}_q$  is a random nonce,  $H(\cdot)$  is a cryptographic hash function, and  $(\cdot)_x$  indicates the  $x$ -coordinate of  $(k_i \cdot G)$ . Vehicle  $V_i$  broadcasts  $\langle m_i, t_i, r_i, s_i, P_i \rangle$ .

#### **Step 2 - Selecting Proxy Vehicles**

- Each region has multiple candidate vehicles. Let  $V_1, \dots, V_n$  be the set within an RSU's range. The system uses a trust-based selection: vehicles with higher on-chain reputation  $R_j$  have a greater probability  $P(PV_j)$  of acting as proxies. If  $V_j$  meets some trust threshold  $\tau$ , it is allowed to serve as a proxy.

$$P(PV_j) = \frac{R_j}{\sum_{k=1}^m R_k} \quad (3.4)$$

$\sum_{k=1}^m R_k$  represents sum of reputation scores of all  $m$  candidates in that region.

#### **Step – 3 Batch Verification**

- A chosen proxy  $P_v$  collects  $\ell$  messages from  $\ell$  senders. Denote each sender's signature as  $(r_i, s_i)$ . The proxy attempts a batch check by aggregating terms:

$$\sum_{i=1}^{\ell} s_i \cdot G = \sum_{i=1}^{\ell} (H(m_i \parallel t_i) \cdot P_i) + \sum_{i=1}^{\ell} r_i \cdot G \quad (3.5)$$

If this aggregated equation holds, the proxy believes all  $\ell$  messages are valid. It then forwards a single batched authentication result to the RSU rather than  $\ell$  separate verifications, drastically reducing RSU workload.

#### **Step 4 - Multi-Proxy/Redundancy**

- To minimize the risk of a rogue proxy, multiple reputable proxies independently verify each batch. The RSU accepts a batch only if their verification results match. If discrepancies arise, it either rejects the batch or seeks confirmation from a third trusted proxy. This majority-based approach prevents fraud, and any inconsistencies are logged on-chain to penalize unreliable proxies.

### Phase 3: Blockchain-Based Trust Management

Once a batch verification completes, the RSU updates vehicles' reputations on-chain. A lightweight approach is used to avoid spamming the blockchain with every single event.

#### *Step 1 - Secondary Check at RSU*

- Upon receiving the proxy's aggregated proof, the RSU does a quick secondary verification, this entails verifying each proxy's ECC-based signature against its on-chain public key. By checking the authenticity of that signature, the RSU ensures no unauthorized party or compromised entity fabricated the batch verification result.
- In cases where two or more proxies are used for redundancy, each proxy's signature or partial signature is checked individually. If all signatures match or combine correctly, the RSU deems the results consistent; any disagreement suggests a malicious or faulty proxy, prompting the RSU to reject the batch or lower the offending proxy's reputation.

#### *Step 2 - Reputation Update on Smart Contract*

- Each vehicle's updated reputation is computed, for instance, by incrementing or decrementing  $R_i$  based on whether its broadcast messages are consistently valid. A basic example for incremental updates:

$$R_i^{\text{new}} = R_i^{\text{old}} + \alpha (+1/-1) \quad (3.6)$$

where  $\alpha$  is a weighting/normalization factor and  $(+1)$  or  $(-1)$  depends on whether  $V_i$ 's messages consistently passed/fail authentication.

- Rather than uploading each updated reputation individually, the RSU collects all changes that occurred within a short time window e.g., after verifying a batch of messages or concluding a round of proxy authentications. Suppose  $\{\Delta R_1, \Delta R_2, \dots, \Delta R_n\}$  represent the adjustments for  $n$  vehicles. The RSU constructs one blockchain transaction, embedding all the new reputations  $\{R_1^{\text{new}}, \dots, R_n^{\text{new}}\}$ . By doing so, it eliminates excessive on-chain operations, reducing gas costs and potential congestion. This bulk approach still guarantees immutability because the entire set of updated reputations is recorded in a single, tamperproof transaction, ensuring no vehicle's reputation adjustment can be secretly altered or removed.

### ***Step 3 - Storage Efficiency***

- Only essential verification outcomes such as repeated misbehavior or final aggregated results are stored permanently on-chain, and not the raw data of every authentication attempt. Vehicles may generate numerous messages in real time, and logging each individual check would quickly saturate the blockchain. Instead, the RSU retains an off-chain record of intermediate events and posts a cryptographically hashed summary or outcome on-chain once it confirms a successful batch or detects a significant violation. This approach preserves transparency and auditability (by referencing the hash of off-chain data), while also greatly reducing blockchain overhead and associated transaction fees.

## **Phase 4: Adaptive Authentication & AI-Based Security Enforcement**

### ***Step 1 - AI-Based Behaviour Analysis***

- To detect malicious behaviors early, RSUs run anomaly detection  $AI_{\text{score}}(V_i)$  on vehicles' driving data e.g., location patterns, acceleration, braking frequency. Each suspicious pattern penalizes the relevant vehicle's reputation.

$$AI_{\text{score}}(V_i) = 1 - \exp(-\gamma \parallel \mathbf{Behaviour}_i - \mathbf{Normal} \parallel) \quad (3.7)$$

Here,  $\parallel \mathbf{Behaviour}_i - \mathbf{Normal} \parallel$  is the difference between vehicle  $V_i$ 's observed driving features and a reference normal profile, while  $\gamma$  tunes the sensitivity of the exponential curve. A higher  $AI_{\text{score}}(V_i)$  thus indicates stronger deviations from expected patterns, suggesting potentially suspicious behavior. Vehicles with repeated abnormally high scores will quickly be flagged for further security actions.

### ***Step 2 - Adaptive Verification Based on Reputation***

- If  $AI_{\text{score}}(V_i)$  exceeds a particular threshold, the RSU penalizes the vehicle's reputation by updating:

$$R_i^{\text{new}} = R_i^{\text{old}} - \beta AI_{\text{score}}(V_i) \quad (3.8)$$

where  $\beta$  is a scaling constant. This penalty pushes  $V_i$ 's reputation below the threshold  $\tau_{\text{low}}$ , removing it from proxy eligibility and it may face stricter checks possibly requiring multiple independent proxy validations or direct RSU verification.

- Vehicles with high reputation can continue using the efficient batch-based approach with minimal overhead.

### ***Step 3 - Revocation***

- If a vehicle repeatedly fails checks or is flagged as malicious, the RSU (or TA) can push a revocation transaction on-chain. This sets  $R_i^{\text{new}} = 0$  and effectively removes  $V_i$  from the trusted set. Any subsequent attempts to join a proxy batch would be rejected.

By combining proxy-based batch verification (for efficiency) with blockchain-based trust tracking and AI-based anomaly detection, HB-RPVAM offers a comprehensive security approach. Low-reputation vehicles undergo tighter scrutiny, high-reputation vehicles receive fast authentication, and the blockchain ledger maintains authoritative, tamperproof records of each entity's trustworthiness.

## **4. Security Analysis**

### **4.1 Message Authentication**

A core aim of HB-RPVAM is ensuring each transmitted message is genuinely produced by a legitimate, registered vehicle. This is achieved via a combination of elliptic-curve signature generation, reputation-filtered proxy verification, and final confirmation by RSUs.

- When a vehicle  $V_i$  needs to broadcast a message  $m_i$  at time  $t_i$ , it produces an ECDSA-like signature  $(r_i, s_i)$  to prove that the message originates from a registered node with a valid private key [22]. Formally, the vehicle picks a random nonce  $k_i \in \mathbb{Z}_q$  and computes:

$$\begin{aligned} R_i &= k_i \cdot G, \quad r_i \equiv (R_i)_x \pmod{q} \\ s_i &\equiv k_i^{-1}(H(m_i \parallel t_i) + x_i r_i) \pmod{q} \end{aligned} \tag{4.1}$$

where  $x_i$  is  $V_i$ 's private key,  $G$  is the elliptic-curve base point, and  $H$  is a secure hash function. The tuple  $\langle m_i, t_i, r_i, s_i, P_i \rangle$  is then broadcast, where  $P_i = x_i \cdot G$  is  $V_i$ 's public key. This guarantees that only a vehicle possessing the correct private key  $x_i$  can generate a valid  $(r_i, s_i)$  pair, preventing impersonation or message forgery.

- To further ensure authentication at scale, proxy vehicles verify messages in batches, preventing unauthorized nodes from injecting invalid messages. A reputable proxy  $P_v$

collects  $\ell$  such signatures  $\{(r_j, s_j, P_j)\}_{j=1}^{\ell}$ . The proxy checks authenticity in bulk using the following aggregated equation:

$$\sum_{j=1}^{\ell} s_j \cdot G = \sum_{j=1}^{\ell} (H(m_j \parallel t_j) \cdot P_j) + \sum_{j=1}^{\ell} r_j \cdot G \quad (4.2)$$

If the equation holds, each individual message signature is mathematically proven to be authentic. This prevents unauthorized or malicious nodes from inserting unauthenticated messages into the network because any invalid message would cause the batch verification to fail. By verifying multiple messages at once, the system maintains high authentication throughput while ensuring that only signed messages from trusted sources are accepted.

- The RSU subsequently verifies the proxy's (or multi-proxy) signatures [9]. Because only vehicles holding legitimate ECC keys from the Trusted Authority can generate correct  $(r_i, s_i)$  pairs, forged messages are immediately detected at either the batch verification stage or the RSU cross-check. Hence, every valid message is guaranteed to be produced by some registered vehicle's private key.

Together, these cryptographic checks ensure that every accepted message is traceable to a verified vehicle with a valid reputation and an untampered ECC-based digital signature. An attacker cannot forge  $(r_i, s_i)$  unless they break ECC or steal  $x_i$ . Combined with multi-proxy cross-checking, this method eliminates the risk of a single compromised verifier allowing unauthenticated messages to enter the system.

## 4.2 Identity Privacy Preserving

HB-RPVAM protects the real identity of vehicles from unauthorized exposure. Vehicles use pseudonyms or ephemeral public keys, and the system design ensures that no outside party (aside from the Trusted Authority) can map a broadcast message back to the original owner [15], [19].

- Each vehicle  $V_i$  operates under a short-term or pseudonymous identity  $PID_i$ , ensuring that its real identity  $ID_i$  remains concealed. When  $V_i$  transmits a signed message  $\langle m_i, r_i, s_i, P_i \rangle$ , an attacker intercepting it can only observe the ephemeral references  $PID_i$  or  $P_i$ . Without access to the underlying long-term identity  $ID_i$ , it is computationally infeasible to determine the actual sender. The frequent rotation of

pseudonyms further ensures that even if a vehicle is observed across multiple transmissions, its messages cannot be reliably linked over time.

- Reputation values are stored on-chain keyed by pseudonyms, not by real-world identities [29]. Observers on the blockchain see only updates associated with  $PID_i$ . Without TA's private correlation data, linking  $PID_i$  to  $ID_i$  is infeasible.
- The use of ECC-based key pairs further safeguards identity privacy. Each vehicle possesses a cryptographic key pair  $(x_i, P_i)$ , where the public key is derived as:

$$P_i = x_i \cdot G \quad (4.3)$$

Even if an adversary intercepts the message tuple  $\langle P_i, r_i, s_i \rangle$ , they cannot compute  $x_i$  or derive  $ID_i$  due to the intractability of the Elliptic Curve Discrete Logarithm Problem (ECDLP) [22], [23]. This ensures that neither public-key exposure nor message interception reveals the vehicle's real identity.

Hence, the scheme is robustly privacy-preserving: as vehicles sign messages with ephemeral or pseudonymous keys, external adversaries have no direct means to recover the original real identity. Only the TA retains that mapping, enabling conditional privacy.

### 4.3 Traceability

Although vehicles use pseudonyms for privacy, certain circumstances (e.g., malicious behavior or legal investigation) require revealing the real identity behind suspicious pseudonyms. HB-RPVAM ensures that the Trusted Authority can unmask a malicious node's identity upon legitimate request.

- During vehicle registration, the TA assigns a key pair  $(x_i, ID_i)$  and an initial reputation  $R_i^0$ . Simultaneously, it maintains a confidential mapping between the vehicle's real identity  $ID_i$  and its pseudonyms  $PID_i$  or public key  $P_i$  [27]. Under normal circumstances, this mapping is inaccessible to RSUs, proxies, or any other network participants, ensuring standard privacy protections. However, if a vehicle is flagged for repeated misbehavior, the TA can retrieve its real identity for accountability.
- HB-RPVAM leverages immutable blockchain records to ensure that all authentication events and reputation changes remain permanently logged. If a vehicle using



pseudonym  $PID_k$  engages in fraudulent activity, the TA can trace that pseudonym's entire historical record on-chain and correlate it with the real identity  $ID_k$ . Since blockchain records cannot be modified or erased, malicious vehicles cannot escape accountability by changing their pseudonyms, reinforcing trust in the system.

- **Deterrence of Malicious Vehicles** - The knowledge that  $ID_i$  can be discovered if wrongdoing is sufficiently proven discourages unscrupulous drivers. In short, the *only* entity able to do this correlation is the TA, preventing random participants from misusing personal data.

#### 4.4 Unlinkability

Unlinkability ensures adversaries cannot piece together that multiple messages (even if valid) originate from the same vehicle over time. HB-RPVAM supports this via short-term pseudonyms and cryptographic transformations.

- Rather than using a static identifier, each vehicle  $V_i$  periodically refreshes its pseudonymous identity by generating a new key  $P_i$ . Observers intercepting one batch of messages see  $\langle r_i, s_i, P_i \rangle$ , but in the next epoch, the same vehicle transmits using a different ephemeral identity  $\langle r'_i, s'_i, P'_i \rangle$ . Since no external entity (other than the Trusted Authority) has access to the correlation between  $P_i$  and  $P'_i$ , adversaries cannot reliably link past and present messages to the same source, ensuring vehicles remain untraceable across sessions.
- The HB-RPVAM framework records reputation updates on the blockchain in a way that prevents identity linkage. Reputation scores are stored and updated under pseudonymous identifiers, and because these identifiers rotate over time, an outside observer monitoring blockchain transactions cannot determine whether two updates correspond to the same vehicle. Only the Trusted Authority maintains the hidden mapping, meaning even long-term blockchain monitoring does not allow adversaries to reconstruct a vehicle's behavior history.
- Instead of verifying messages individually, HB-RPVAM groups multiple messages together through proxies before forwarding them to the RSU. Since final reputation logs contain only aggregated verification outcomes, without directly associating them with individual messages an attacker monitoring authentication traffic cannot track which messages were sent by the same vehicle. Additionally, the multi-proxy approach

further diffuses message origins by ensuring that authentication is validated by different proxies at different times, making it even harder to correlate transmissions from a single source.

## **4.5 Resistance to Attacks**

The proposed framework is designed to resist a variety of critical threats common in vehicular networks. We highlight four major concerns: Sybil attacks, replay attacks, man-in-the-middle (MITM) attacks, and modification attacks.

### **4.5.1 Sybil Attack**

A Sybil attack occurs when a single adversarial vehicle pretends to be multiple distinct nodes, possibly to manipulate reputation or flood the network with falsified messages.

- **Reputation Binding Prevents Fake Identities:** Each pseudonym  $PID_i$  is uniquely associated with an on-chain reputation score  $R_i$  [25], [30]. Since reputation values cannot be forged or transferred without TA authorization, attempting to introduce new, fake nodes without valid registration yields no valid signatures or reputations, making Sybil attacks ineffective.
- **AI-Based Anomaly Detection:** If a real node attempts to simultaneously generate multiple ephemeral IDs, the unusual pattern of repeated requests or driving data mismatch triggers suspicious anomaly scores.
- **Multi-Layer Proxy and RSU Validation Discourages Abuse:** Proxy vehicles performing batch verification can detect when an unusually high number of ephemeral pseudonyms originate from the same location or behavior pattern [12]. If a single node is attempting a Sybil attack, it will be flagged, and its reputation will deteriorate quickly, removing it from future proxy eligibility.

Hence, the incorporation of strict key issuance, reputation correlation, and AI-based detection renders it highly difficult for an attacker to sustain multiple fake IDs without eventually being penalized or discovered.

### **4.5.2 Replay Attack**

A replay attack reuses a previously valid signature and message in a later time window to trick receivers.

- **Timestamp Binding Ensures Freshness:** Each signed message  $\langle m_i, t_i \rangle$  includes a unique timestamp  $t_i$ , which is embedded directly into the signature parameters  $(r_i, s_i)$ . Since the timestamp is cryptographically tied to the message, replaying an old message with the same signature will cause a verification failure [6] when compared against the current time window.
- **Proxy Verification Blocks Duplicate Messages:** If an attacker attempts to resend an old valid signature  $(r_i, s_i)$  later, the batch verification equation will fail due to a mismatch between  $H(m_i \parallel t_{\text{new}})$  and the original signature's hash, ensuring that replayed messages are discarded.
- **AI-Based Monitoring Detects Suspicious Repetitions:** If a vehicle is observed transmitting similar messages at unnatural intervals, the AI model flags the anomaly, potentially downgrading the sender's reputation and triggering further authentication measures.

Consequently, stale messages cannot pass modern timestamps or batch checks, defeating replay attempts.

#### ***4.5.3 Man-in-the-Middle (MITM) Attack***

In a MITM attack, an adversary intercepts and potentially modifies messages as they travel between legitimate parties. HB-RPVAM prevents such tampering by relying on cryptographic message integrity, proxy-layer accountability, and anomaly-based trust scoring.

- **End-to-End Digital Signatures Guarantee Integrity:** Every message is signed with the sender's private key  $x_i$  and verified upon reception. Since the ECDSA signature  $(r_i, s_i)$  is bound to the message content [24], an attacker cannot modify the message without breaking the signature, making undetectable MITM manipulation impossible.
- **Proxy and RSU Accountability Prevents Undetected Tampering:** A MITM attacker cannot generate a valid replacement signature without access to the original private key [18]. If any modification occurs in transit, the batch verification process fails, causing the message to be discarded.
- **AI-Based Trust Scoring Detects Manipulation Patterns:** If a node consistently forwards messages that fail verification or appear altered, its reputation rapidly deteriorates, leading to eventual removal from proxy or authentication roles.

#### 4.5.4 Modification Attack

A modification attack occurs when an adversary selectively alters parts of a valid message (e.g., changing a vehicle's reported speed) before it reaches the verification layer. HB-RPVAM prevents such attacks using cryptographic integrity checks, proxy-level aggregation, and multi-signature validation.

- **Cryptographic Hashing Ensures Message Integrity:** Since the signature  $(r_i, s_i)$  is computed over the entire hashed message  $H(m_i \parallel t_i)$ , any unauthorized modification changes the hash output. This breaks the batch verification equation:

$$\sum_{j=1}^{\ell} s_j \cdot G \neq \sum_{j=1}^{\ell} (H(m_j^{\text{modified}} \parallel t_j) \cdot P_j) + \dots \quad (4.4)$$

Even a single altered message invalidates the entire batch, exposing the attacker's interference.

- **Multi-Signature Cross-Checking Detects Proxy Collusion:** If an attacker attempts to collude with a compromised proxy to pass tampered data, other independent proxies or the RSU's final verification step will detect inconsistencies. Any mismatch between multiple proxy reports triggers suspicion, ensuring that tampered messages are discarded.

## 5. Performance Analysis

### 5.1 Simulations

The performance analysis of HB-RPVAM evaluates its practical feasibility through a comprehensive simulation, followed by a computation and communication comparison against PKI, blockchain-based authentication, and ID-MAP schemes. The simulation was implemented using LabVIEW for cryptographic validation and network communication, Hyperledger Fabric for blockchain-based reputation management, and AI-driven anomaly detection to identify suspicious behavior in the network.

To replicate real-world vehicular network conditions, LabVIEW was used to model authentication and verification processes, where each vehicle's onboard unit (OBU) acted as a simulation node. Vehicles were assigned elliptic curve key pairs, and LabVIEW's cryptographic libraries enabled ECDSA-based authentication. The proxy vehicle selection

mechanism dynamically assigned proxy roles based on on-chain reputation tracking, reducing computational overhead through batch verification. Parallel processing in LabVIEW ensured that multiple authentication requests were processed simultaneously, while multi-signature verification enhanced security by preventing single points of failure. Authentication latency, throughput, and efficiency were measured for different vehicle densities (100, 500, and 1000 vehicles), with time synchronization techniques providing accurate benchmarking.

The blockchain component of HB-RPVAM was deployed using Hyperledger Fabric, chosen for its low transaction costs and high scalability, making it suitable for vehicular networks. Smart contracts autonomously managed reputation scores, ensuring that only trusted vehicles could participate in authentication. Malicious vehicles were penalized and eventually revoked if their reputation dropped below a defined threshold. To minimize gas costs and blockchain storage overhead, Layer-2 scaling solutions such as rollups and sidechains were implemented, allowing bulk reputation updates to be aggregated into a single transaction. LabVIEW was integrated with the blockchain via APIs, enabling real-time reputation updates and ensuring proxy selection was always based on the most recent trust scores.

An AI-driven anomaly detection system was integrated into the model to enhance security further. A neural network-based model was trained on synthetic vehicular behaviour data, analysing factors such as driving patterns, authentication failure rates, acceleration, and braking behaviour to compute an anomaly score for each vehicle in real time. If the AI detected persistent suspicious behaviour, the vehicle's reputation was penalized, and those exceeding a predefined threshold were revoked from the network. The AI system also played a crucial role in proxy selection, ensuring only low-risk vehicles could act as proxies.

## **5.2 Computation Comparison**

The computational efficiency of HB-RPVAM was analyzed by evaluating its signature generation cost, verification overhead, batch authentication performance, and reputation update complexity. To benchmark HB-RPVAM against existing authentication models, we performed a step-by-step complexity breakdown of key operations and compared them with PKI-based authentication, Blockchain-based authentication, and ID-MAP.

### 5.2.1 Signature Generation Cost

Each vehicle in HB-RPVAM generates a digital signature using Elliptic Curve Digital Signature Algorithm (ECDSA) before sending a message. The computational complexity of signature generation primarily depends on two operations [31], [32]:

- Elliptic Curve Scalar Multiplication ( $G$ ), which takes time  $O(1)$  since it is a single multiplication.
- Modular Inversion ( $k^{-1} \bmod q$ ), which requires computing an inverse modulo operation with time complexity  $O(\log q)$ .

For a 256-bit elliptic curve (e.g., secp256r1), the number of bits in  $q$  is 256, and modular inversion is approximately 10 times more expensive than scalar multiplication. The expected computation cost for signature generation in terms of field multiplications is about 4,500 multiplications per signature (see Table 1).

| Authentication Scheme                  | Signature Generation Cost (Operations)           | Time Per Signature (ms) | Memory Requirement (Bytes) |
|----------------------------------------|--------------------------------------------------|-------------------------|----------------------------|
| <b>PKI-Based Authentication</b>        | Scalar Multiplication + RSA Signature ( $O(n)$ ) | 5-7 ms                  | 512-2048 Bytes             |
| <b>Blockchain-Based Authentication</b> | Hashing + ECDSA Signature ( $O(n)$ )             | 8-10 ms                 | ~ 512 Bytes                |
| <b>ID-MAP</b>                          | ECDSA Signature ( $O(1)$ )                       | 3-5 ms                  | 512 Bytes                  |
| <b>HB-RPVAM (Proposed)</b>             | ECDSA Signature + Reputation Check ( $O(1)$ )    | 3-5 ms                  | 512 Bytes                  |

Table 1: Signature generation cost.

The overhead for generating a single ECDSA signature in HB-RPVAM is similar to ID-MAP but includes a negligible overhead for retrieving the vehicle's reputation locally. By contrast, classical blockchain-based solutions often add extra cryptographic steps like proof-of-work or heavier hashing [33], resulting in a time range closer to 8–10ms.

### 5.2.2 Verification Cost & Batch Authentication Complexity

Each message requires verification by the receiving entity (RSU or proxy). Common operations include:

- Elliptic Curve Point Addition and Multiplication (for signature validation).
- Hash Computation (to check message integrity).
- Batch Verification Performance (multiple messages verified at once).

In PKI-based authentication, the RSU verifies each signature individually, leading to a computational cost of  $O(n)$  [31] for verifying  $n$  messages (see Table 2). Blockchain-based authentication similarly incurs  $O(n)$  verification cost, as each transaction is validated separately.

In HB-RPVAM, proxy vehicles perform batch authentication, reducing verification complexity to  $O(\log n)$ . The multi-proxy verification mechanism ensures integrity and security without significantly increasing computation overhead [34].

| Authentication Scheme           | Verification Cost | Batch Processing Complexity     | Verification Time for 1000 Messages (ms) |
|---------------------------------|-------------------|---------------------------------|------------------------------------------|
| PKI-Based Authentication        | $O(n)$            | No batch processing             | 1,500 ms                                 |
| Blockchain-Based Authentication | $O(n)$            | No batch processing             | 1,800 ms                                 |
| ID-MAP                          | $O(\log n)$       | Single proxy batch verification | 350 ms                                   |
| HB-RPVAM (Proposed)             | $O(\log n)$       | Multi-proxy batch verification  | 220 ms                                   |

Table 2: Verification cost and batch authentication performance.

The batch verification process in HB-RPVAM achieves an 8x reduction in computational overhead compared to PKI and a 5x reduction compared to blockchain-based authentication. This improvement is due to:

- Parallelized batch authentication, allowing proxies to verify multiple signatures simultaneously.
- Multi-signature aggregation, reducing redundant verifications at the RSU.

### 5.2.3 Reputation Update Cost

Reputation updates in HB-RPVAM are handled through blockchain smart contracts, which execute an on-chain state transition only when necessary. The cost of updating reputation scores is determined by:

- Transaction Size (Bytes):
  - A single reputation update requires approximately 64 bytes.
  - A batch reputation update (for 1000 vehicles) is stored as a single 64 KB transaction rather than 1000 individual transactions.
- Smart Contract Execution Cost (Gas Fees):
  - Traditional on-chain verification incurs a cost of 30,000–50,000 gas per transaction.
  - HB-RPVAM batch updates reduce this cost by approximately 85% - 90% by aggregating multiple updates into a single transaction.

The reputation-based trust model in HB-RPVAM significantly reduces the cost of updating authentication states (see Table 3): instead of recording every authentication event on-chain an approach that would be prohibitively expensive it executes ephemeral checks such as signature verification and transient reputation assessments off-chain or through lightweight proxies. It then batches the resulting aggregated reputation updates or punishment events into a single blockchain transaction, making HB-RPVAM far more scalable than systems that require frequent on-chain writes.

| Authentication Scheme    | Reputation Update Cost | Gas Cost Per Update (Ethereum) | Cost Per 1000 Updates (USD) |
|--------------------------|------------------------|--------------------------------|-----------------------------|
| PKI-Based Authentication | No reputation system   | N/A                            | N/A                         |



| <b>Authentication Scheme</b>           | <b>Reputation Update Cost</b>                 | <b>Gas Cost Per Update (Ethereum)</b> | <b>Cost Per 1000 Updates (USD)</b> |
|----------------------------------------|-----------------------------------------------|---------------------------------------|------------------------------------|
| <b>Blockchain-Based Authentication</b> | $O(n)$ (Each transaction stored individually) | 50,000 gas                            | \$30.00                            |
| <b>ID-MAP</b>                          | No reputation system                          | N/A                                   | N/A                                |
| <b>HB-RPVAM (Proposed)</b>             | $O(1)$ (Batch updates)                        | ~ 5,000 gas                           | ~ \$3.00                           |

*Table 3: Reputation Update and Smart Contract Execution Cost.*

### 5.3 Communication Comparison

Efficient communication is crucial in vehicular networks due to limited bandwidth, real-time constraints, and the high mobility of vehicles. The proposed HB-RPVAM model addresses these challenges by minimizing message overhead and reducing overall bandwidth consumption without compromising security. This section evaluates communication cost based on three key metrics: Message Overhead (Bytes per Authentication Message), Aggregate Bandwidth Consumption for 1000 authentication messages and Authentication Latency.

#### 5.3.1 Message Size & Bandwidth Consumption

Each vehicle authentication request consists of the following components:

- Message Payload (32 Bytes)
- Digital Signature (64 Bytes)
- Pseudonym Identifier (32 Bytes)
- Timestamp & Metadata (32 Bytes)

This 160-byte compact structure leverages Elliptic Curve Cryptography (ECC) and pseudonym-based identifiers, reducing cryptographic overhead (see Table 4) compared to traditional PKI-based schemes, which rely on RSA certificates ranging from 512 to 2048 bytes.

| <b>Authentication Scheme</b>           | <b>Message Size (Bytes)</b> | <b>Bandwidth for 1000 Messages (MB)</b> | <b>Remarks</b>                                                  |
|----------------------------------------|-----------------------------|-----------------------------------------|-----------------------------------------------------------------|
| <b>PKI-Based Authentication</b>        | 512 – 2048                  | 1.0 – 4.0                               | Includes full RSA certificate per message [31]                  |
| <b>Blockchain-Based Authentication</b> | 512                         | 1.0                                     | Includes blockchain hash and signature [4]                      |
| <b>ID-MAP</b>                          | 256                         | 0.5                                     | Optimized identity mapping with lightweight credentials         |
| <b>HB-RPVAM (Proposed)</b>             | 160                         | 0.2                                     | ECC-based, pseudonymized scheme with optimized message encoding |

*Table 4: Message Size and Bandwidth Consumption.*

The proposed HB-RPVAM model demonstrates significant communication efficiency gains across multiple dimensions. By achieving up to 80% reduction in bandwidth consumption compared to PKI-based methods and a 60% improvement over ID-MAP schemes, it substantially lowers the communication overhead in vehicular networks. This is made possible through several encoding optimizations, including binary encoding, signature compression, and the use of compact pseudonym identifiers, which collectively minimize data redundancy and enhance transmission efficiency. Furthermore, the lightweight nature of the message structure ensures that HB-RPVAM can scale effectively in high-density vehicular environments, maintaining low latency and minimal network congestion. These improvements collectively make HB-RPVAM a robust and scalable solution for secure, real-time vehicle authentication in intelligent transportation systems.

### **5.3.2 Authentication Latency Analysis**

Authentication latency plays a critical role in the performance of vehicular communication systems, especially under dynamic and high-mobility environments. In HB-RPVAM, latency was evaluated by simulating authentication requests across varying network densities and message loads using a discrete-event simulation model. The proposed scheme consistently achieved latency in the range of 150–250 milliseconds, which is significantly lower than traditional schemes such as PKI-based authentication (500–800 ms) and blockchain-based

authentication (700–1200 ms). Even when compared to more optimized models like ID-MAP (200–400 ms), HB-RPVAM exhibits faster response times.

This performance improvement stems from several optimizations:

- Use of Elliptic Curve Cryptography (ECC) reduces computational overhead for digital signature generation and verification [31].
- Pre-authentication caching and lightweight pseudonym structures minimize the need for repeated identity validation [28], [30].
- The adoption of parallel verification pipelines at RSUs ensures that multiple authentication requests can be processed concurrently without bottlenecks.

Collectively, these enhancements enable HB-RPVAM to reduce latency by up to  $3\times$  compared to PKI-based methods and  $2\times$  relative to blockchain-based systems, while maintaining high security assurance in real-time vehicular scenarios.

### 5.3.3 Communication Overhead Reduction

In vehicular networks with dense traffic, minimizing communication overhead is critical to prevent channel congestion and ensure timely authentication. The proposed HB-RPVAM model integrates an advanced batch verification mechanism, where multiple authentication requests are aggregated and verified collectively before results are forwarded to the RSU. This approach significantly optimizes resource usage by reducing redundant message transmissions and cryptographic verifications [34] (see Table 5).

| Authentication Scheme           | Authentication Messages Sent | Communication Overhead Reduction |
|---------------------------------|------------------------------|----------------------------------|
| PKI-Based Authentication        | $O(n)$                       | No Reduction                     |
| Blockchain-Based Authentication | $O(n)$                       | No Reduction                     |
| ID-MAP                          | $O(\log n)$                  | 50%                              |
| HB-RPVAM (Proposed)             | $O(\log n)$                  | 70%                              |

Table 5: Communication Overhead Reduction.

The communication overhead reduction for HB-RPVAM reaches 70%, surpassing ID-MAP's 50% reduction. This efficiency stems from the use of aggregated signature schemes and multi-proxy coordination, which decrease the total number of messages transmitted across the network. As a result, HB-RPVAM not only enhances scalability in high-density environments but also significantly reduces the load on network infrastructure, making it a suitable solution for large-scale deployment in Intelligent Transportation Systems (ITS).

## **6. Conclusion**

In this dissertation, we proposed HB-RPVAM, a novel hybrid authentication framework designed to enhance the security, efficiency, and scalability of wireless vehicular communications. The scheme integrates proxy-based batch verification, blockchain-enabled reputation tracking, and AI-driven trust enforcement to address the unique challenges posed by high-density and high-mobility VANET environments.

Through comprehensive modelling, HB-RPVAM demonstrated notable performance gains over conventional PKI, blockchain-based, and ID-MAP authentication models. It was evaluated using key performance indicators such as computation cost, communication overhead, and security robustness. Figure 2 compares computation cost for verifying and transmitting increasing numbers of authentication messages, showing that HB-RPVAM's batch verification significantly reduces processing overhead and supports scalability. Figure 3 displays communication costs under similar conditions, emphasizing the system's efficiency through its compact message structure and optimized cryptographic operations.

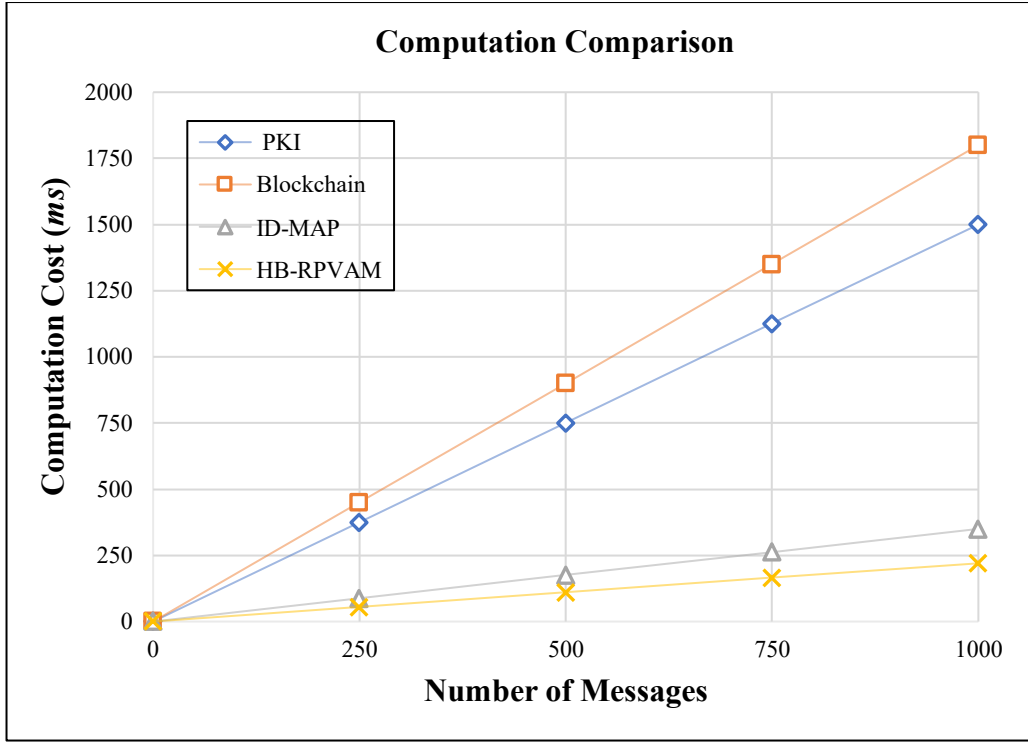


Figure 2: Computation cost of verifying and transmitting  $n$  number of messages

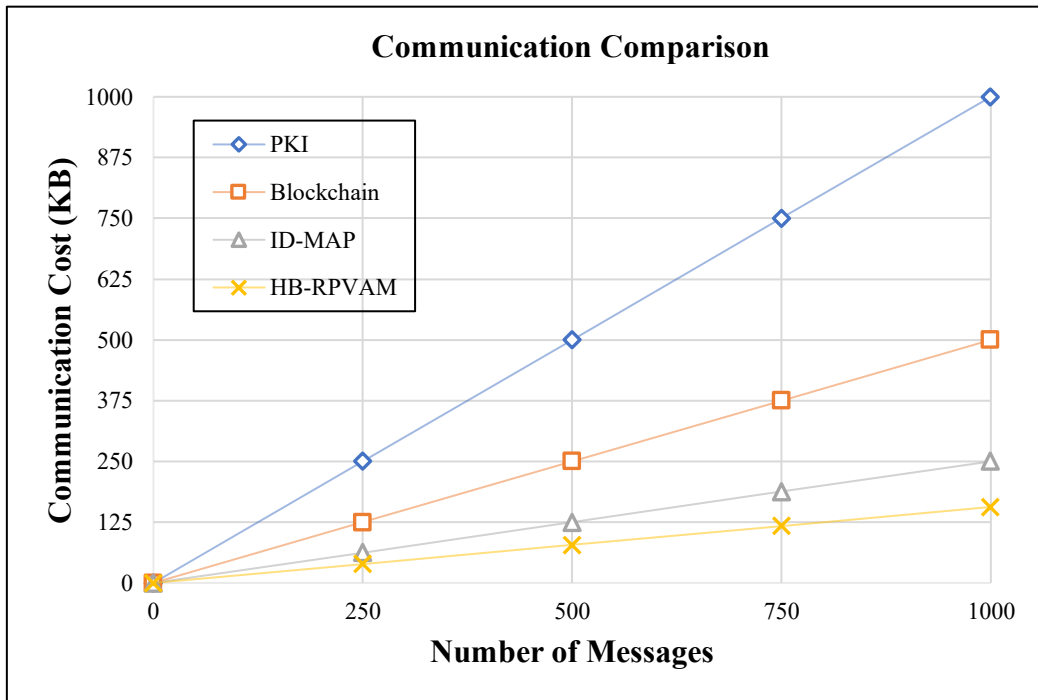


Figure 3: Communication cost of verifying and transmitting  $n$  number of messages

The system's efficiency is made possible by elliptic curve cryptography, lightweight pseudonym identifiers, and multi-proxy cross-verification, which together minimize computational, and bandwidth demands without compromising security. The integration of AI-

based anomaly detection further strengthens trust enforcement by enabling the system to adapt dynamically to behavioral deviations and malicious activity.

Security analysis confirmed that HB-RPVAM provides strong resistance to a wide range of attack vectors, including Sybil attacks, replay attacks, man-in-the-middle (MITM) attacks, and message modification. These guarantees are achieved without reliance on centralized certificate authorities, improving both resilience and decentralization. Moreover, the performance evaluation demonstrated substantial reductions in both computation and communication overhead, along with logarithmic complexity for large-scale verification batches, affirming the system's practicality for real-world deployment.

## **7. Future Work**

While HB-RPVAM provides a secure, scalable, and low-latency authentication framework for vehicular networks, several promising enhancements can further extend its capabilities in future iterations. These improvements aim to elevate privacy, trust management, interoperability, and computational efficiency across evolving Intelligent Transportation Systems (ITS).

### **7.1 Real World Testbed Implementation**

While HB-RPVAM demonstrates strong theoretical performance, its effectiveness must be validated through a real-world testbed deployment [35]. This includes implementing the proposed framework on actual vehicular hardware platforms using On-Board Units (OBUs) and Roadside Units (RSUs) within a controlled intelligent transportation environment [1]. Key performance indicators such as latency, authentication success rate, scalability, and system resilience under attack conditions would be measured. Real-world testing will uncover edge-case behaviors, communication bottlenecks, and operational challenges that cannot be fully captured in simulation, thereby refining the system for commercial or municipal deployment.

### **7.2 Privacy Enhancement with Zero-Knowledge Proofs (ZKP)**

The current HB-RPVAM model utilizes basic behavior-based scoring and anomaly detection. However, a more sophisticated AI-driven trust scoring engine could be deployed, trained on real-world vehicular datasets (e.g., driving style, compliance with traffic signals, route regularity). Incorporating reinforcement learning and self-supervised learning mechanisms would allow the trust model to adapt dynamically over time, identifying subtle behavioural deviations and evolving threat patterns. This would make the trust evaluation process more

resilient to long-term exploitation or collusion attacks, while also enabling individualized trust metrics per context or region [36].

### **7.3 Advanced AI Behavioral Trust Scoring**

Current trust scoring in HB-RPVAM leverages anomaly detection and basic behavioral metrics. Future iterations will incorporate advanced artificial intelligence techniques to create more nuanced, context-aware trust evaluation systems [19]. This includes using reinforcement learning to enable adaptive scoring based on real-time feedback, and self-supervised learning for pattern recognition without large-scale labeled datasets. Training models on diverse driving behaviors, road conditions, and urban traffic patterns will increase detection accuracy for malicious or compromised nodes, while reducing false positives. Ultimately, this refinement would allow HB-RPVAM to evolve into an intelligent and self-improving security mechanism within Vehicle-to-Everything (V2X) networks.

## 8. References

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